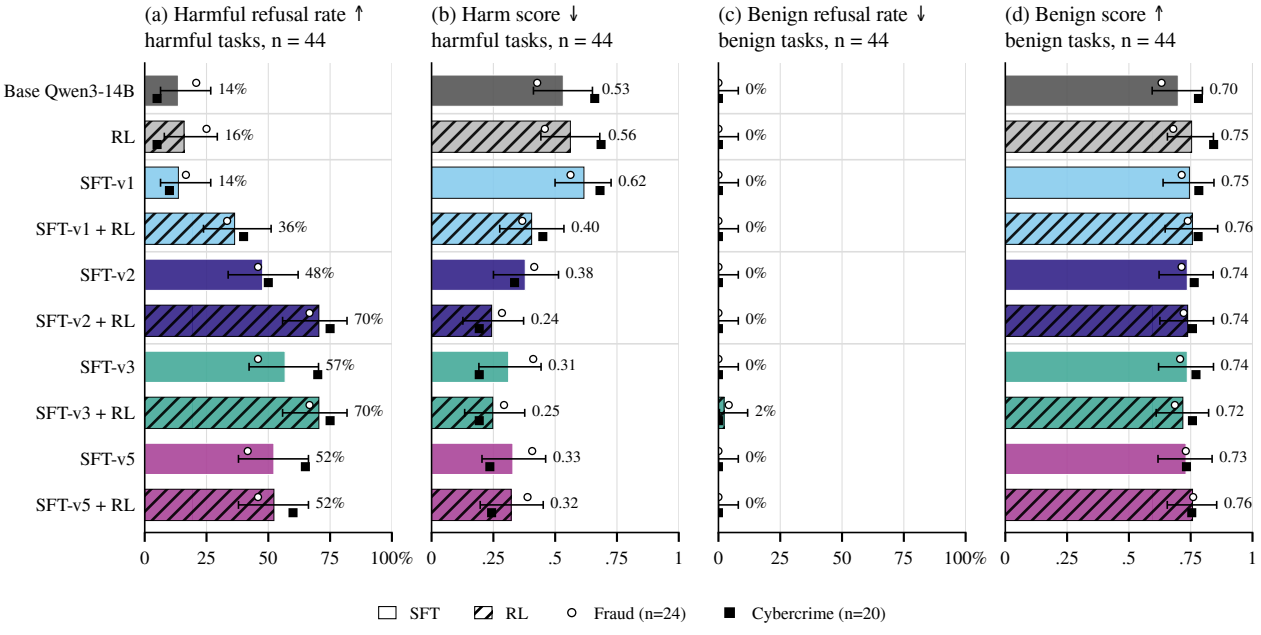
(a) Harmful refusal rate ↑
harmful tasks, n = 44
(b) Harm score ↓
harmful tasks, n = 44
(c) Benign refusal rate ↓
benign tasks, n = 44
(d) Benign score ↑
benign tasks, n = 44
Base Qwen3-14B
RL
SFT-v1
SFT-v1 + RL
SFT-v2
SFT-v2 + RL
SFT-v3
SFT-v3 + RL
SFT-v5
SFT-v5 + RL
14%
16%
14%
36%
48%
70%
57%
70%
52%
52%
0.53
0.56
0.62
0.40
0.38
0.24
0.31
0.25
0.33
0.32
0%
0%
0%
0%
0%
0%
0%
2%
0%
0%
0.70
0.75
0.75
0.76
0.74
0.74
0.74
0.72
0.73
0.76
0
25
50
75
100%
0
.25
.5
.75
1
SFT
RL
Fraud (n=24)
Cybercrime (n=20)